

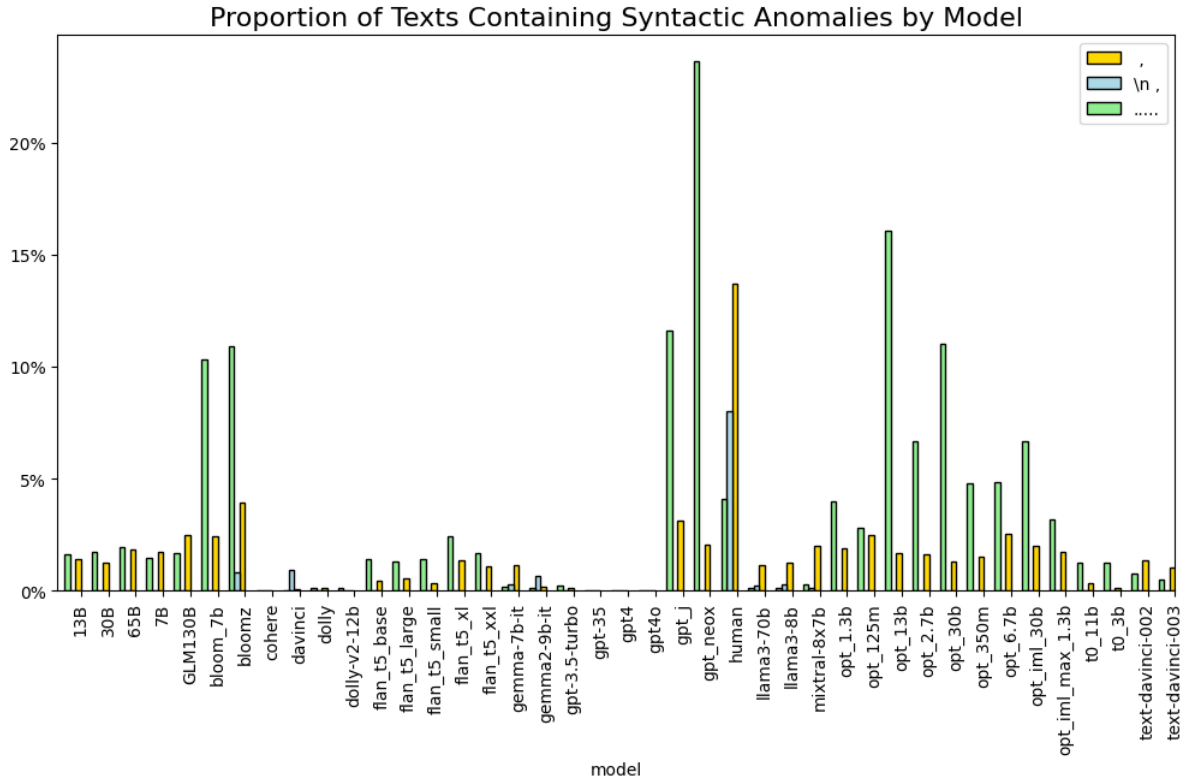


Figure 9: Frequency of occurrence of three common syntactic anomalies - spaces before commas, commas after line breaks, and ellipses with more than three dots in the text samples generated by different models. The vertical axis represents the percentage of COLING dataset samples in which each anomaly appears at least once, while the horizontal axis indicates the generation models.

feature 1416), structural (e.g., sentence endings, introduction words, numbering, feature 6587), and discourse-level (e.g., concept flow, reformulations, contradictions, features 4645, 8689). Manual inspection of documents with high feature values offers complementary interpretative insights. Below, we overview the most expressive features.

General features. (Table 3)

According to steering-based explanation, all presented features makes text lengthy and overwinded, but with different flavour: feature 3608 increases sentence complexity, feature 4645 responsible for knowledge presentation complexity (even without real knowledge), and feature 6587 incorages lengthy introductionds and explanations. According the manual analysis, the first of them is concentrated on “scientifically-looking” tokens, the second reacts on factual contradictions, and the third is activated in structural elements of the text, like item labels or introduction words.

GPT-specific features.

In Table 4 we present features detecting well modern LLMs, especially GPT family. Feature 8689 responsible for excessive synonym substitu-

tions, and feature 8264 for thoughts repetitions (by steering interpretation); from the examples we can see that the first is activated on paraphrased ideas already mentioned in the text, or on discussing alternatives. The second is activated on long common words, specific for typical GPT style.

Domain-specific features.(Tables 5, 6)

Feature 12390 (arxiv) is responsible for syntactic complexity. It is activated linking structures typical for scientific writing.

Feature 1416 (wikihow) is interpreted as increasing “phylosofical or metaphorical explanations” instead of being simple and clear. In fact, its extreme values succesfully detects texts where crucial parts are missing, namely, results of parsing errors where formulas and mathematical characters are lost. So, discarding mathematical characters is the extreme case of the unclarity.

Feature 6513 (finance) represent exsessive explanations behind clear facts. It is activated on opinionate words and syntactic constructions “I mean”, “like” etc

Feature 14953 (medicine) responsible for second-person speech with direct instructions. Ac-

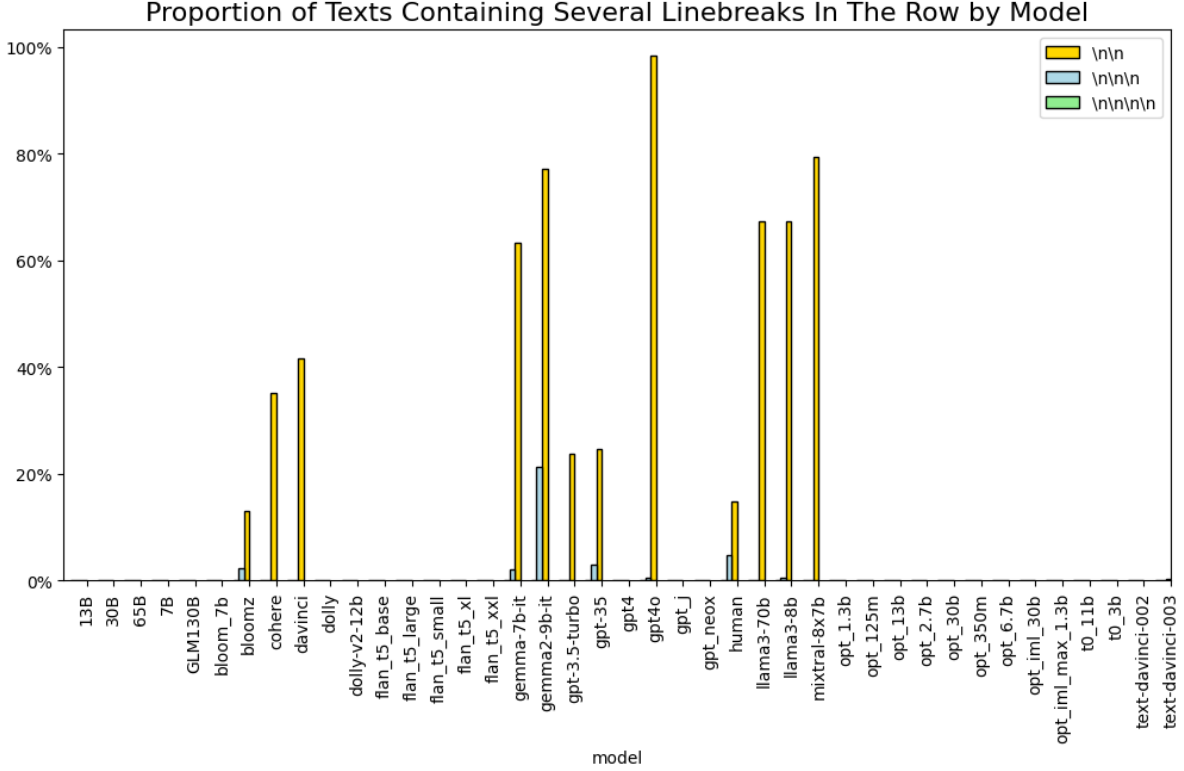


Figure 10: Frequency of occurrence of the excessive line breaks - namely, two, three or four line breaks in the row. The vertical axis represents the percentage of COLING dataset samples in which each amount of excessive line breaks appears at least once, while the horizontal axis indicates the generation models.

tivated on phrases containing “You” or “Your” pronouns. Steering interprets it as change from informal to formal language.

Feature 4560 (reddit) responsible for “speculative causality”, which Reddit discussions as its extreme implementation

Feature 4773 reacts on words flexibility. Steering interprets it as “hallucinations”.

G Steering: additional details

Feature steering was applied using shifts from the following set: $\{-4.0, -3.0, -2.5, -2.0, -1.5, -1.0, -0.5, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0\}$. To analyze the effects of these modifications, we utilized the GPT-4o model. The prompt is shown in Figure 17.

H Effect of steering

In Table 7, we present examples of steering for several features with their GPT-based interpretations using three prompts. While GPT generally captures the influence of the features, some effects are not fully accounted for. For instance, feature 6513 causes unnecessary expansion in factual questions

(prompt P2) but adds positive intent in opinionated contexts. Feature 4773 enhances writing sophistication, feature 1416 boosts creativity in fictional contexts and causes hallucinations in factual ones, and feature 14953 turns every response into legal advice.

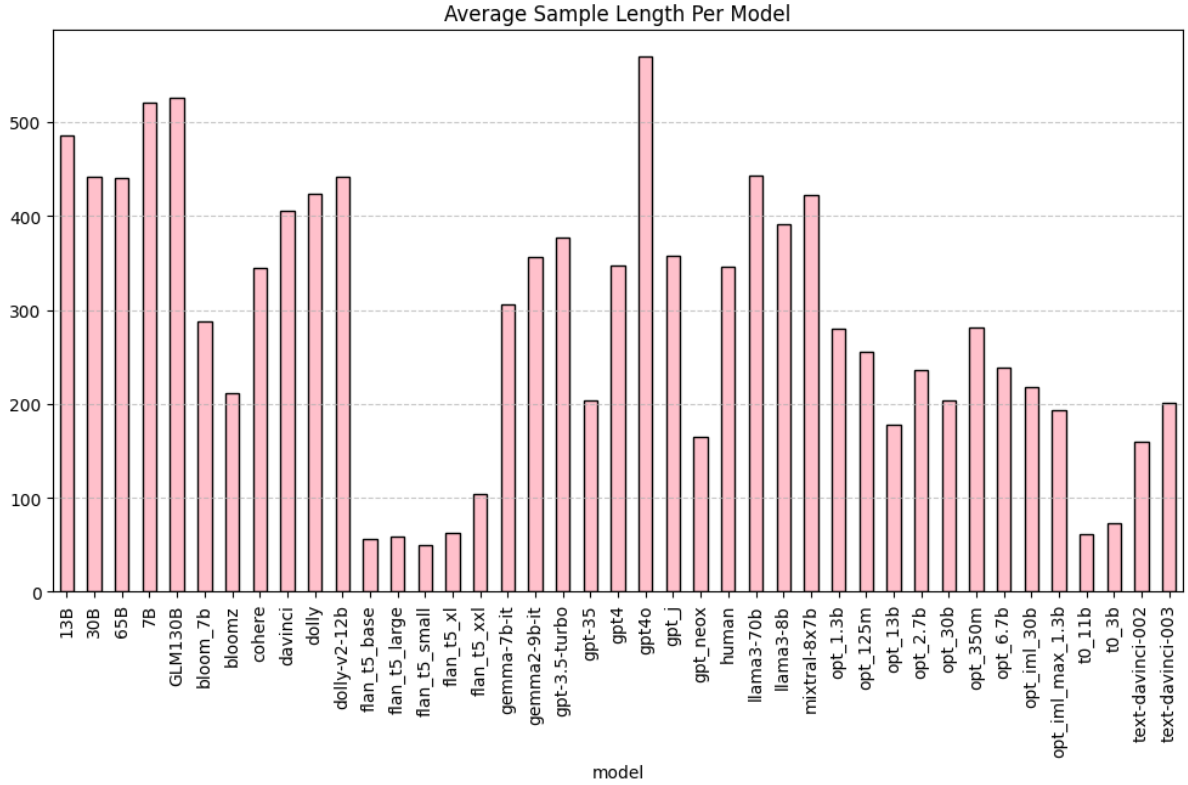


Figure 11: Average length of the text sample in COLING dataset by the generation model. The vertical axis represent the text length (measured in Gemma-2-2B tokens), the horizontal axis indicates the generation models.

This paper presents a comprehensive study on multiple and single snapshot compressive beamforming, a technique used in signal processing and array processing. The study explores the theoretical underpinnings of the method, its applications, and its limitations. The paper also compares the performance of multiple snapshot compressive beamforming with single snapshot compressive beamforming. The results indicate that multiple snapshot compressive beamforming provides superior performance in terms of resolution and noise suppression. However, it also requires more computational resources. The paper concludes with suggestions for future research and potential improvements in the technique.

Figure 12: GPT-4 generation, "misspelling" attack